

Exam #2- CHEM 330- Fall 2025

Name..... - ANSWERS

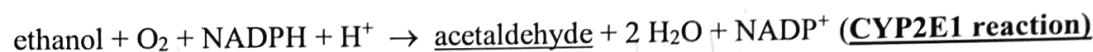
A) Multiple Choice Section (30 min). 24 questions. Only one correct answer. 2.5 points each.

1) In the first step of ethanol metabolism in the liver, ethanol is oxidized to acetaldehyde through the action of alcohol dehydrogenase (ADH):



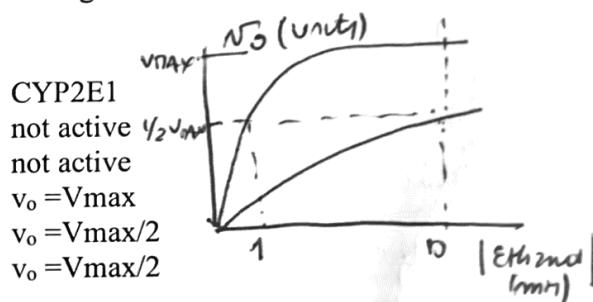
(ADH reaction)

Ethanol can be also oxidized to acetaldehyde through a minor pathway catalyzed by an enzyme called cytochrome P450 2E1 (CYP2E1). This enzyme catalyzes the following reaction:

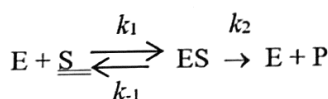


The K_m for ethanol oxidation by CYP2E1 is 10 mM; the K_m for ethanol oxidation by ADH is 1 mM. Assume that V_{max} for both CYP2E1 and ADH is the same. After drinking two beer cans, the approximate blood ethanol concentration in a standard 70 kg male is 10 mM. Under these conditions:

- ADH
- A) not active
 - B) $v_o = V_{max}/2$
 - C) $v_o = V_{max}$
 - D) $v_o = V_{max}/2$
 - E) $v_o = V_{max}$**



2) Michaelis and Menten assumed that the overall reaction for an enzyme-catalyzed reaction could be written as

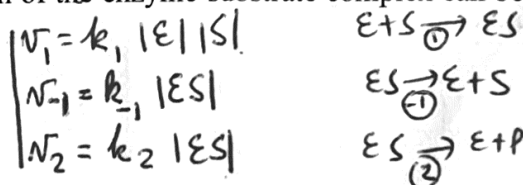


Using this reaction, the rate of breakdown of the enzyme-substrate complex can be described by the expression:

- A) $k_1 ([E_t] - [ES])$
- B) $k_1 ([E_t] - [ES])[S]$
- C) $k_2 [ES]$
- D) $k_{-1} [ES] + k_2 [ES]$**
- E) $k_{-1} [ES]$

In general, for an elementary reaction:
 $A \rightarrow B$
 $N = -\frac{d[A]}{dt} = -\frac{d[B]}{dt} = k[A]^1$

So, for a sequence of three elementary reactions:



$$\begin{aligned} \text{So, } N_{\text{breakdown ES}} &= N_{-1} + N_2 \\ &= k_{-1} [ES] + k_2 [ES] \end{aligned}$$

Relevance: You might have heard two expressions: "The liver has a constant capacity to eliminate alcohol from circulation of 1 drink/hour," and "Limit your alcohol intake." ① is justified because ADH is almost always acting at V_{max} = constant - ② is justified because only at high [ethanol] is CYP2E1 metabolism important - CYP2E1 uses O_2 in its mechanism, which creates toxic reactive oxygen species as side-products -

67% wrong

3) The initial velocity assumption for Michaelis-Menton kinetic assumes that _____

- A) the reaction is always running at V_{max} .
- ☒ B) the rate of $E + P \rightarrow ES$ is negligible.
- C) a large amount of product has formed.
- D) k_{-2} is large.
- E) none of the above.

Reaction -2 is neglected before
 $t \approx 0$ since I am working at
 $t \approx 0$, when P has not accumulated
 yet-

4) The kinetic data in the table below were obtained using an enzyme known to follow Michaelis-Menten kinetics:

[S] mM	0.8	2.0	4.0	6.0	500.0	1000.0
Velocity (micromol per min)	217	325	433	488	635	647

$\rightarrow$ 1/2 that gives $1/2 V_{max} = 325$

$1/2 V_{max} \approx 325$

325 $\mu\text{mol/min}$

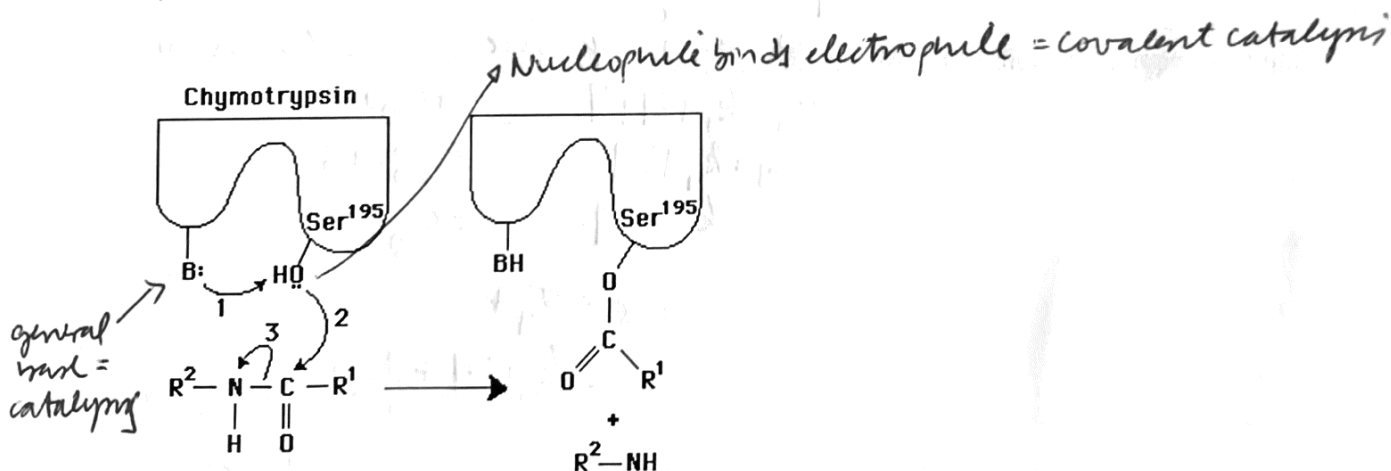
$V_{max} \approx 650 \mu\text{mol/min}$

The K_m for the substrate of this enzyme is approximately:

- A) 325 micromol/min
- B) 650 micromol/min
- ☒ C) 2 mM
- D) 4 mM
- E) 500 mM

5) In the following diagram of the first step in the reaction catalyzed by the protease chymotrypsin, the process of general base catalysis is illustrated by the number _____, and the process of covalent catalysis is illustrated by the number _____.

- ☒ A) 1; 2
- B) 1; 3
- C) 2; 3
- D) 2; 1
- E) 3; 2

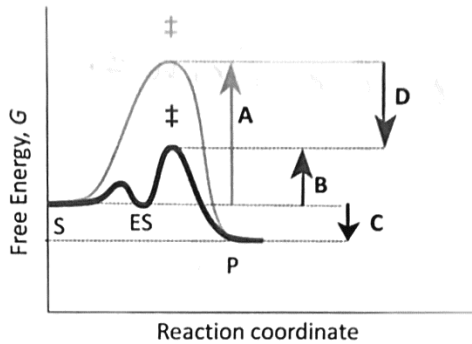


6) Which of the 'change in free energy' arrows in the diagram below represents 'binding energy'?

81%
wrong

- A) A
- B) B
- C) C
- (D) D

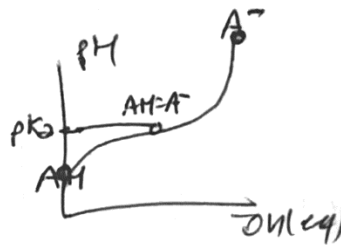
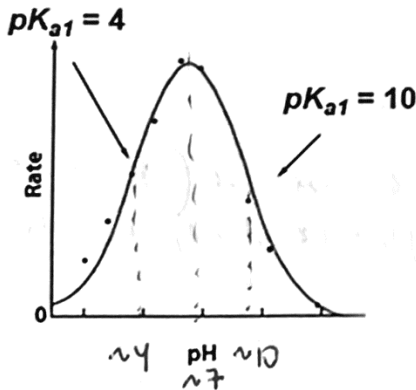
E) none of the arrows represent "binding energy"



7) Based on the plot shown (rate = k_{cat}), which of the following statements is NOT true?

59%
wrong

- A) The optimal pH of this enzyme would be around 7 ✓
- B) The residue with a pKa of 4 is likely acting as a general base in catalysis at the optimal pH. ✓
- (C) At the optimal pH, the residue with a pKa of 10 is in its conjugate base form.
- D) The residue with a pKa of 4 could be a Asp or Glu residue *As, typical pKa*
- E) All of the above are true statements



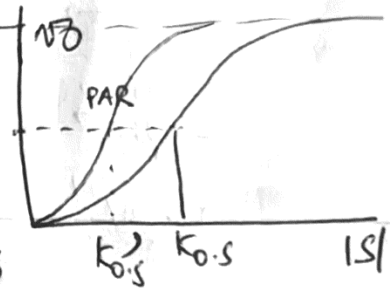
$pH 7 > pK_{a1}$
 A^- (base) predominates -

$pH 7 < pK_{a2}$, AH (acid) predominates - X

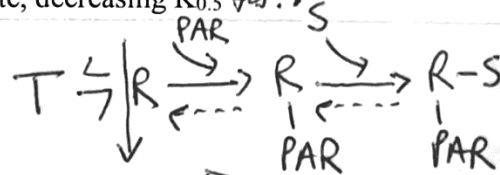
8) AMP is a positive allosteric regulator of the enzyme phosphofructokinase-1. It:

33%
wrong

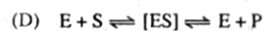
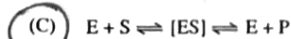
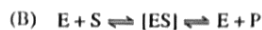
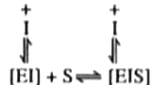
- A) Increases the number of T conformations *opposite*
- B) Causes a shift to the right in the sigmoid V_0 vs $[S]$ curve *opposite*
- C) Stabilizes the T state, increasing $K_{0.5}$ *opposite*
- D) Decreases the cooperativity of the substrate
- (E) Stabilizes the R state, decreasing $K_{0.5}$ *As!*



④ Cooperativity: binding of the substrate to one active site increases the affinity of other active sites, because the enzyme shifts from T (low affinity) to R (high affinity) -



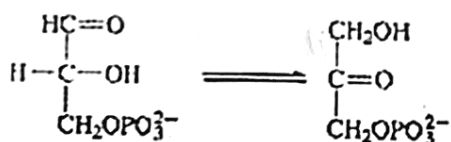
9) Which of the following diagrams best describes the equilibria associated with a competitive inhibitor (I)?



binds reversibly to the active site in E.

10) The reaction below is catalyzed by an enzyme belonging to which **class of enzymes**?

- 51%
wrong
- A) Hydrolase
 - B) Oxidoreductase
 - ☒ C) Isomerase
 - D) Ligase
 - E) Transferase



these 2 are isomers (glyoxaldehyde and dihydroxyacetone, phosphorylated form).

11) Which of the following side chains is least likely to function as a nucleophilic catalyst?

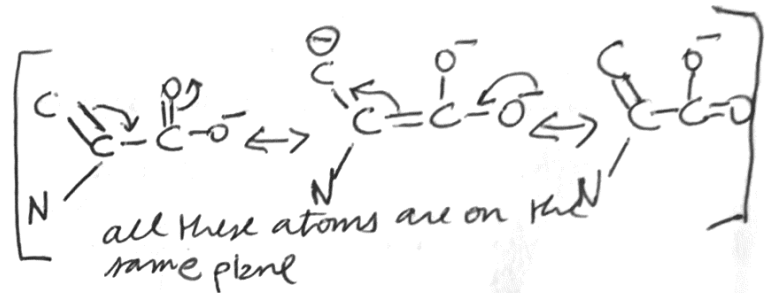
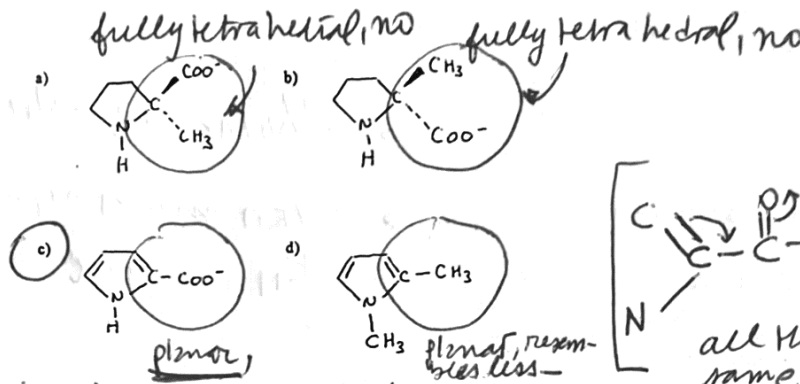
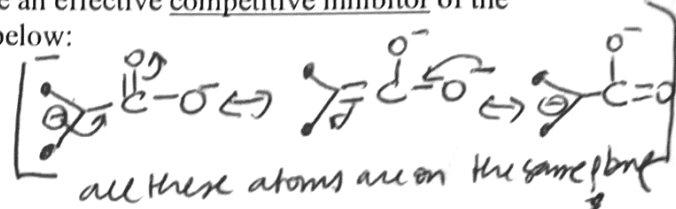
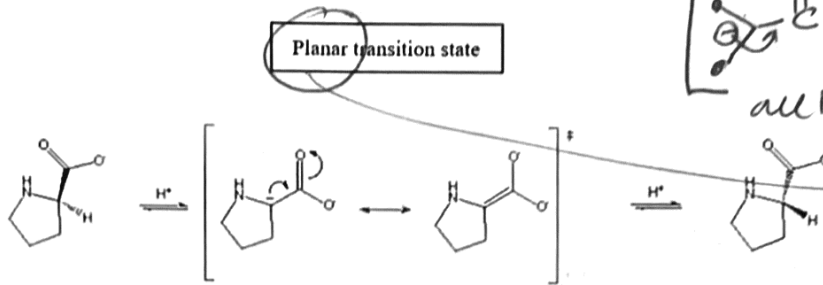
- A) cysteine $-\text{SH}$
- B) serine $-\text{OH}$
- C) histidine $-\text{N}^{\delta-}$
- ☒ D) alanine $-\text{CH}_3$
- E) tyrosine $-\text{OH}$

-CH₃: no nucleophile

43%
wrong

12) Which of the following structures is most likely to be an effective competitive inhibitor of the enzyme proline racemase, which catalyzes the reaction below:

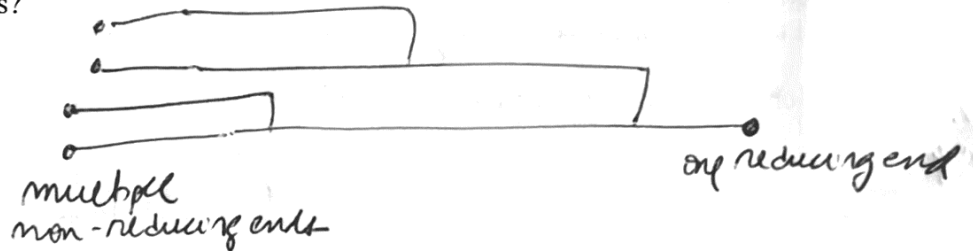
= transition state analogs
are the best competitive inhibitors -



46%
wrong

13) How many reducing ends are in a molecule of glycogen that contains 10,000 residues with a branch every 10 residues?

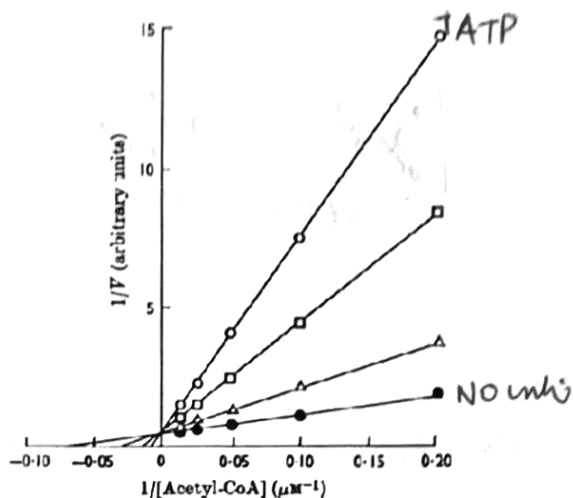
- A) 10,000
- B) 1,000
- C) 100
- D) 10
- E) 1



14) Which of the following statements about starch and glycogen is false?

- A) Amylose is unbranched; amylopectin and glycogen contain many ($\alpha 1 \rightarrow 6$) branches. True
- B) Both are homopolymers of glucose. true
- C) Both serve primarily as structural elements in cell walls. NO storage
- D) Both have ($\alpha 1 \rightarrow 4$) glycosidic bonds. True
- E) Glycogen is more extensively branched than starch. True

15) The enzyme citrate synthase catalyzes the biosynthesis of citrate using acetyl-coA and oxaloacetate as substrates. The biomolecules (adenylate nucleotides) ATP, ADP and AMP were found to be inhibitors of citrate synthase, as shown in the following plots:



- No adenylate nucleotide □ 4 mM ADP
 ○ 4 mM ATP △ 4 mM AMP

Complete the following sentence using the options below:

ATP is a(n) -----inhibitor with respect to acetyl-coA.

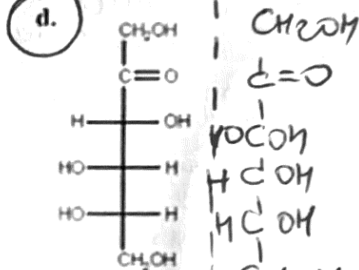
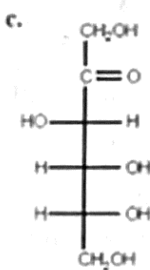
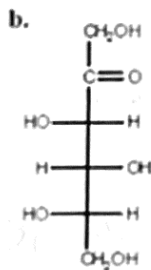
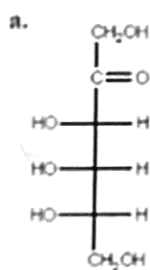
- A) Allosteric
 B) Competitive
 C) Uncompetitive
 D) Mixed
 E) Pure noncompetitive

→ $= 1/V_{MAX}, 20, = V_{MAX}$

→ $-1/K_M$ more right, so K_M increased

68%
wrong

16) Which of the following represents L-fructose?



CH₂OH
 C=O
 HO-C-OH
 H-C-OH
 H-C-OH
 CH₂OH
 D-fructose

17) You are testing several enzyme variants that catalyze the same reaction. The kinetic parameters are shown below:

Enzyme	k_{cat} (s^{-1})	K_m (μM)
A	50	50
B	50	5
C	500	50
D	500 <i>highest k_{cat}</i>	5 <i>lowest K_m</i>
E	100	10

catalytic efficiency: $\uparrow k_{cat}$ (high catalysis)
 $\downarrow K_m$ (high binding)

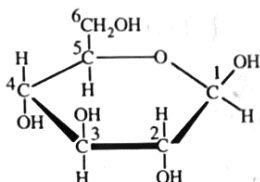
$$= k_{cat}/K_m$$

$$cat. efficiency = 500/5 units = 10 units$$

Which enzyme would be expected to **catalyze the reaction most efficiently?**

- A) Enzyme A
- B) Enzyme B
- C) Enzyme C
- D) Enzyme D**
- E) Enzyme E

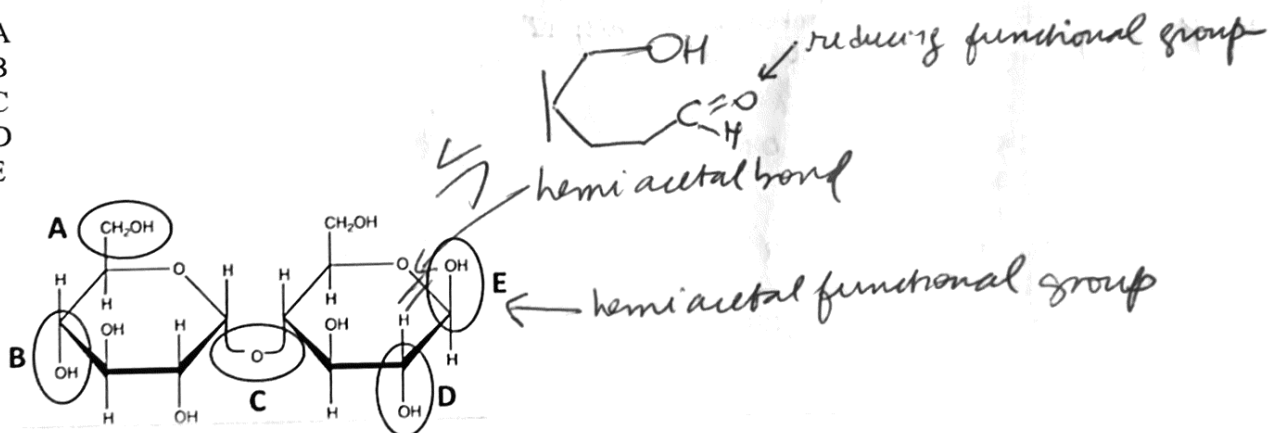
18) In the following compound, carbon 1 (i) is the anomeric C, and carbons 1-5 (ii) are chiral.



- A) i=6 and ii= 2,3,4,5
- B) i=2 and ii= 3,4,5,6
- C) i=1 and ii= 1,2,3,4,5**
- D) i=5 and ii=6
- E) i=1 and ii=1,2,3,4,5,6

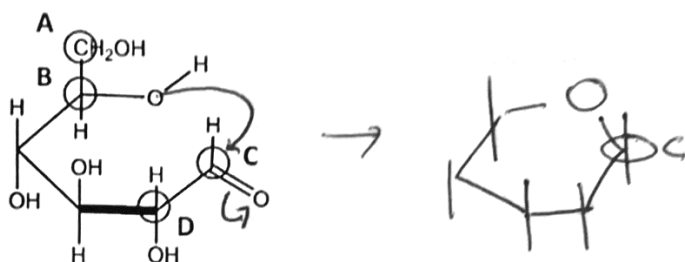
19) Which chemical group determines that this is a 'reducing' disaccharide?

- A) A
- B) B
- C) C
- D) D
- E) E**



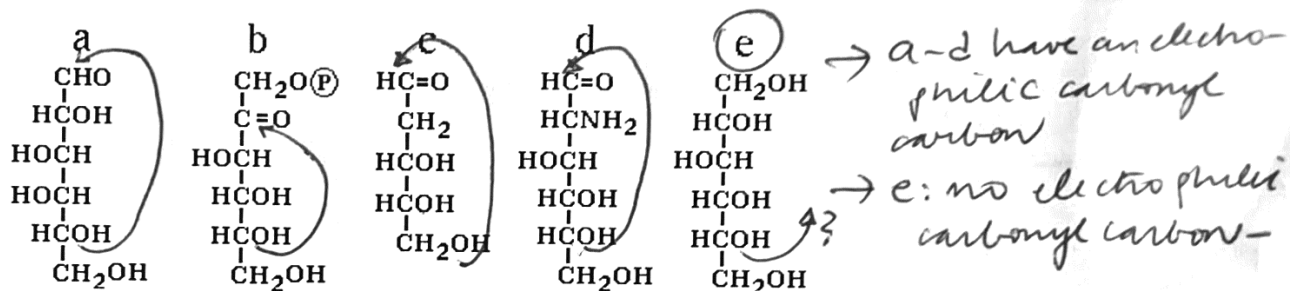
20) In the glucose molecules shown, which of the circled carbons will become the anomeric carbon in the cyclic structure?

- A) A
B) B
C) C
D) D



21) Which of the following sugars cannot exist in a cyclic form? Note: "CHO" represents an aldehyde group.

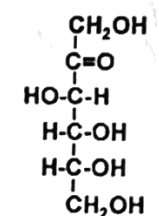
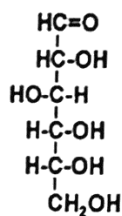
65%
wrong



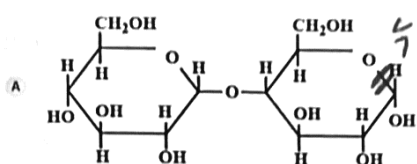
22) Which of the following best describes the following two structures?

- A) They are C2 epimers NO, aldose/ketose pair, structural isomers-
B) They are both aldose sugars NO, one aldose, one ketose
C) They are anomers NO! linear configuration
D) They can form hemiketal bonds only IV
E) Can be interconverted by an isomerase Yes!

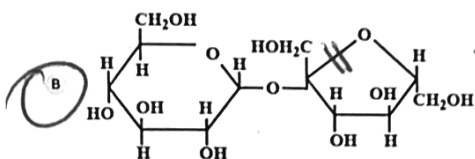
35%
wrong



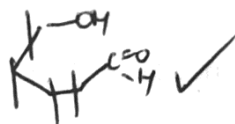
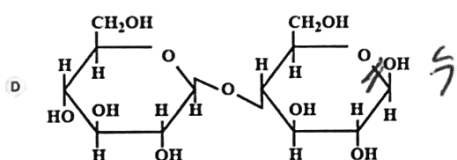
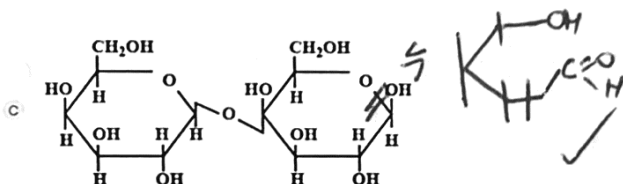
23) Which of the following disaccharides is a non-reducing sugar?



break the hemi acetal or hemi ketal bond. If you can't regenerate a free aldehyde group, then it's non-reducing

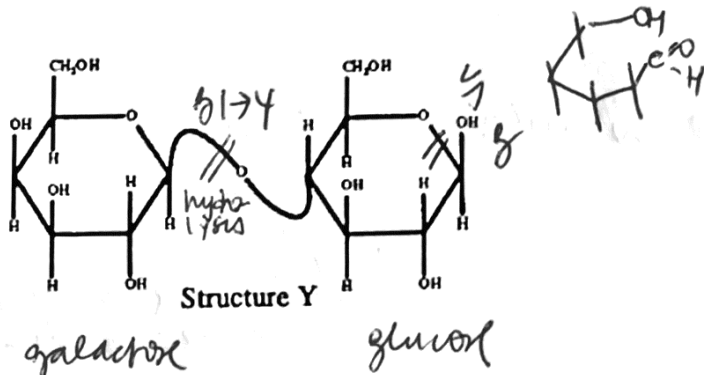


(careful, structure on the right is flipped; anyway, you discarded all the rest) -



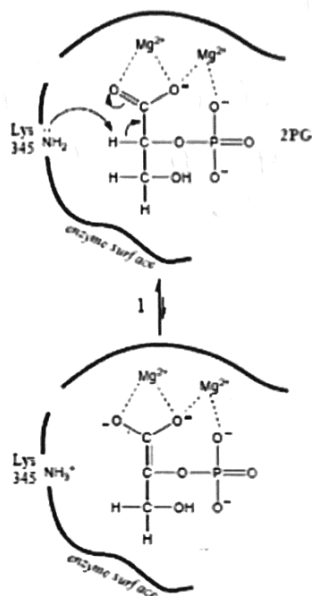
24) The only false statement (out of A, B, C, D and E) below concerning Y is _____

- A) Y is a reducing sugar because it has one free anomeric carbon atom ✓
- B) Y has a $(\beta 1 \rightarrow 4)$ glycosidic linkage and has β -configuration on its free anomeric carbon ✓
- C) On hydrolysis, one mole of Y will yield one mole each of glucose and galactose (a C4 epimer of glucose) ✓
- D) Y is not a reducing sugar because all its anomeric carbon atoms are involved in glycosidic bonds (no, only one on the left) -
- E) Y does not contain a furanose ring in its structure ✓ True, all pyranoses -



B) Short Answers Section (20 min). 4 questions. 10 points each

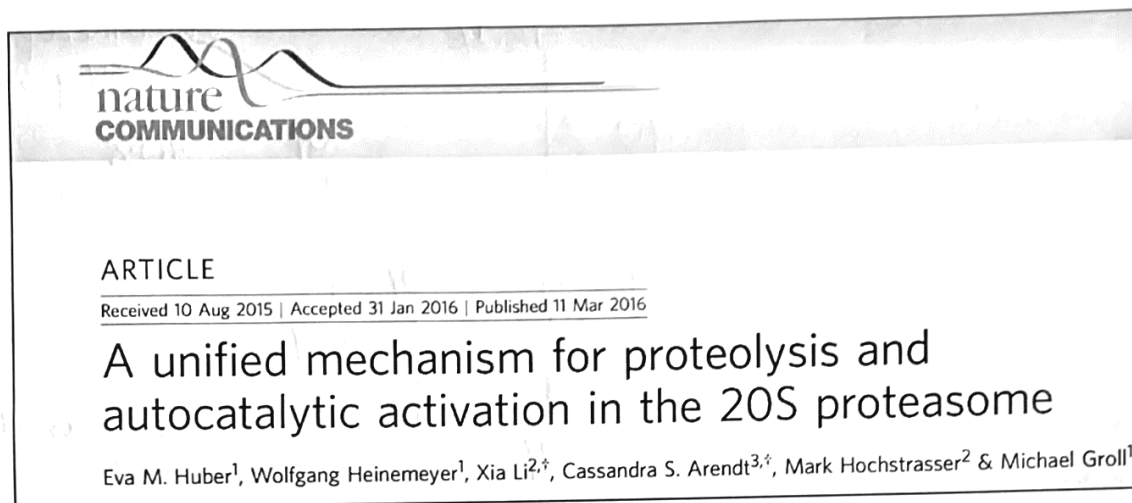
1) a) The enzyme enolase catalyzes the conversion of 2-phosphoglycerate (2PG) into phosphoenolpyruvate (PEP), an important step in the breakdown of sugars. The mechanism of the first step in the enolase reaction is shown below:



Consider the components involved in the enolase mechanism, including the active-site residue Lys345 and the Mg^{2+} cofactors, and complete the table below:

Component	Catalytic mechanism (name)	Catalytic mechanism (describe)
Lys ³⁴⁵	general base catalysis	The basic amino group of lysine removes a proton from the substrate
Mg^{2+}	Metal-ion catalysis	- Mg^{2+} orients the substrate in the right position in the active site - Mg^{2+} polarizes a $\text{C}=\text{O}$ bond, making the carbonyl carbon more electrophilic - Mg^{2+} stabilizes a negatively charged intermediate

b) In 2004, the protease field was honored by a Nobel Prize in Chemistry presented to Avram Hershko, Aaron Ciechanover, and Irwin Rose for their pioneering work on the proteasome system, in which the proteasome is a novel threonine protease. Very recently, a new mechanism for proteolysis in the proteasome was proposed by a multinational group:

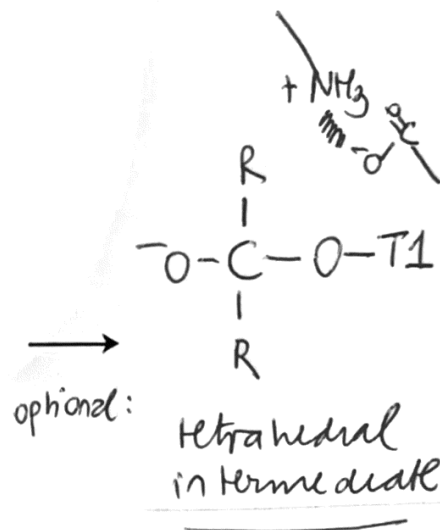
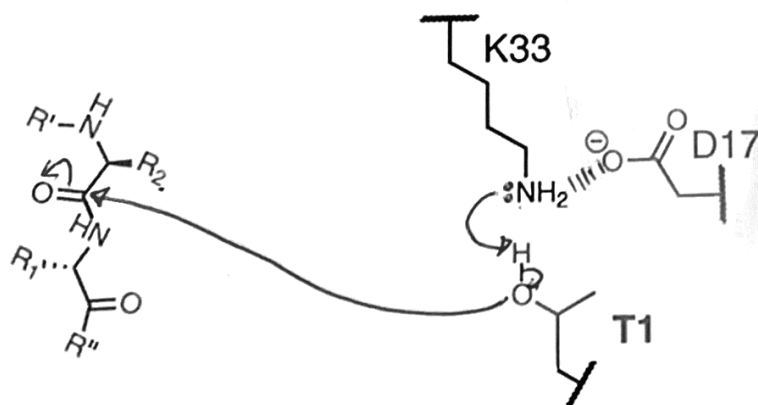


“On the basis of these results, we propose that the proteasome from all domains of life uses a catalytic triad consisting of Thr1, Lys33, and Asp17 for proteolysis”.

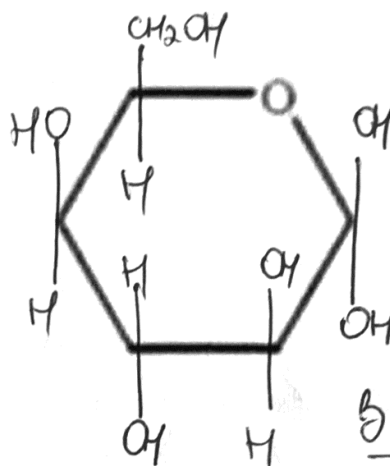
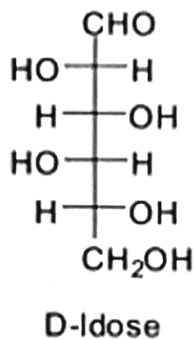
- Thr1 acts as the **nucleophile**. → *If there is a nucleophile, where is the electrophile?*
- Lys33 acts as the **base** that activates Thr1.
- Asp17 **stabilizes the protonated Lys33**.

Complete the mechanism (shown below) for the first step of catalysis in the active site of the proteasome. Use **curly arrows** to show the movement of electron pairs.

Residues: **T1 (Thr1)**, **K33 (Lys33)**, **D17 (Asp17)**. The peptide bond to be cleaved is on the left.



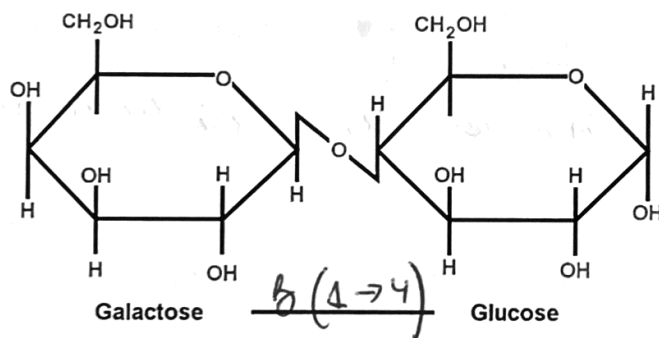
2) a) i) Below is the open-chain structure of D-idose, an aldohexose. a) On the backbone provided, draw the cyclic form. ii) What is the full name of the sugar you have drawn?



Incorrect position
of -OH groups:
30%.

β -D-idopyranose

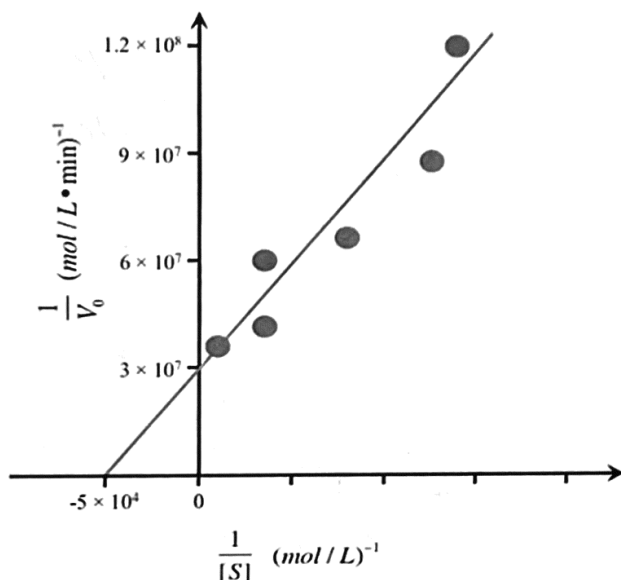
b) i) Name the disaccharide drawn below:



3) a) You measure the initial velocity (V_0) of an enzyme at six different substrate concentrations, then plot these data on a Lineweaver-Burk plot (below). Calculate the Michaelis-Menten parameters for this enzyme. Show your work.

- a) The V_{max} of the enzyme is: $(1/3 \times 10^7) \text{ M/min}$
 b) while the K_m of the enzyme is $(\frac{1}{5} \times 10^4) \text{ M}$

- The student reports that $V_{max} = 3 \times 10^7$ and $K_m = 5 \times 10^4$ or the student confuses this graph with a saturation curve; 35%.



$$y \text{ intercept} = \frac{1}{V_{max}}$$

$$x \text{ intercept} = -1/K_m$$

$$\text{So, } V_{max} = \frac{1}{y \text{ intercept}}$$

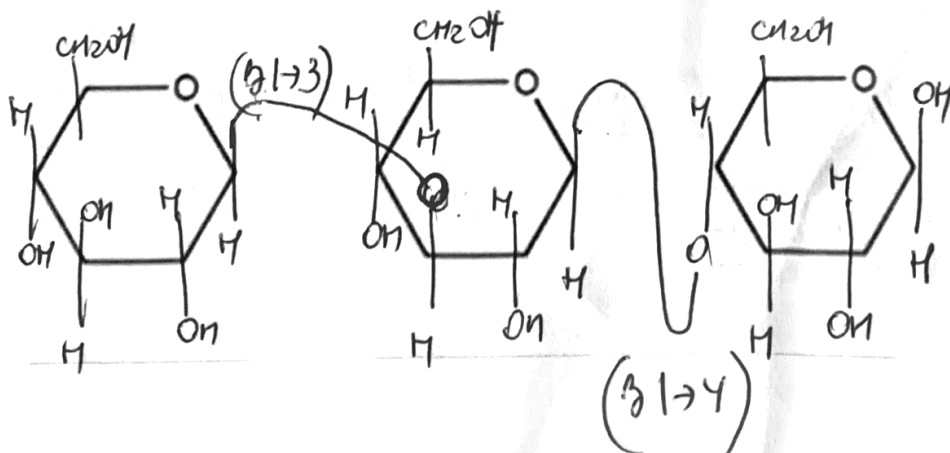
$$K_m = \frac{-1}{x \text{ intercept}}$$

[Caful; $\frac{1}{3 \times 10^7} = \frac{1}{3} \times \frac{1}{10^7} = 0.33 \times 10^{-7}$
 - Didn't take off points because of this -]

b) You are a biochemist studying unusual carbohydrates found in rare rainforest plants. You have isolated a polysaccharide that contains alternating $\beta(1 \rightarrow 4)$ and $\beta(1 \rightarrow 3)$ linkages between D-glucose residues.

On the framework provided:

- Complete each cyclic β -D-glucopyranose structure.
- Connect the first two glucose units with a $\beta(1 \rightarrow 4)$ O-glycosidic bond.
- Connect the last two units with a $\beta(1 \rightarrow 3)$ O-glycosidic bond.
- Clearly label each glycosidic linkage ($\beta(1 \rightarrow 4)$ and $\beta(1 \rightarrow 3)$) on your drawing.



4) For each polysaccharide listed below (cellulose, amylose, amylopectin, and glycogen), indicate Yes or No for each characteristic in the table.

	Cellulose	Amylose	Amylopectin	Glycogen
β 1 $\rightarrow$ 4 bonds	Yes	No	No	No
α 1 $\rightarrow$ 4 bonds	No	Yes	Yes	Yes
α 1 $\rightarrow$ 6 bonds	No	No	Yes	Yes
Helical	No	Yes	Yes	Yes
Storage polysaccharide	No	Yes	Yes	Yes

Amino acid	α -COOH pK_a	α -NH ₃ ⁺ pK_a	R-group pK_a
Gly	2.3	9.6	-
Ala	2.4	9.7	-
Val	2.3	9.6	-
Leu	2.4	9.6	-
Iso	2.4	9.7	-
Met	2.4	9.2	-
Pro	2.1	10.6	-
Phe	1.8	9.1	-
Trp	2.4	9.4	-
Ser	2.2	9.2	13
Thr	2.6	10.4	13
Tyr	2.2	9.1	10.1
Cys	1.7	10.8	8.3
Asn	2.0	8.8	-
Gln	2.2	9.1	-
Asp	2.1	9.8	3.9
Glu	2.2	8.8	4.3
Lys	2.2	9.0	10.5
Arg	2.2	9.0	12.5
His	2.4	9.2	6.0